TEACHER GUIDE · GRADE 3 · DEBUGGING

The Wrong Block Inside

Quick facilitation notes & answer key

Goal

Students find a loop where the repeat count is right but an inside block is wrong (C3.2), in block and turtle contexts, and describe how the wrong block changes the result (a square becomes a line).

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app and/or Python Turtle projected; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Run the clap, clap loop; ask 'what was it supposed to do?' (clap, stomp).
2. Run the forward, forward square — it draws a line; the turn block is missing.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: check every inside block, not just the count.

Answer key (checked by the run-gate)

Ex 1 — the second block is wrong; change clap to stomp.

Ex 2 — the second forward is wrong; change it to bit.right(90).

Ex 3 — spin.

Ex 4 (challenge) — with two forwards and no turn, Bit just keeps going straight (8 forwards); the right(90) turn block is missing.

Watch for & differentiate

Common slip: fixing the repeat count when the count is fine and a block is wrong.

Support: have students say each inside block aloud and compare to the goal.

Extend: ask which block is wrong if a triangle loop has forward, forward (the turn).

Success looks like

The student locates the wrong inside block (not the count) and replaces it so the result matches the goal (Ex 1–3), and explains why a missing turn makes a line (Ex 4).